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Toward Reliable and Intelligent Sensor Systems: A Comprehensive Study of Fault Diagnosis and Mitigation

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ABSTRACT Sensors are central elements of modern engineering systems, supporting essential functions in areas, such as healthcare, manufacturing, autonomous systems, environmental monitoring, and smart infrastructure. As sensor networks become larger and more integrated into intelligent and networked environments, ensuring their reliability has become increasingly challenging. Faults such as bias, drift, and signal loss can severely degrade data integrity, system performance, and safety. This article offers a comprehensive review of sensor fault management, going beyond the scope of previous surveys by addressing the full lifecycle, from fault detection and diagnosis (FDD) to mitigation and compensation. While most existing reviews tend to focus primarily on detection and diagnosis, this work takes a step further by examining a range of advanced mitigation strategies, including hardware redundancy, analytical reconstruction, artificial intelligence-based approaches, and improved calibration and maintenance practices. The review also provides a structured classification of fault types, investigates their causes and effects across different application areas, and critically evaluates the strengths, assumptions, and limitations of current FDD methods. By bringing together a broad, cross-domain perspective, and a detailed analysis of existing methods, this work aims to be a useful reference for researchers and engineers working on reliable, fault-tolerant, and intelligent sensor systems. It also highlights several ongoing challenges, such as separating actual faults from noise, addressing cybersecurity risks, and ensuring that sensor networks can scale and adapt to the demands of complex, safety-critical environments.

INDEX TERMS Advanced calibration and maintenance, analytical compensation, data quality, degrade system performance, fault decoupling, fault mitigation, lifecycle of sensor fault management, sensor fault diagnosis (SFD), sensors.

I. INTRODUCTION

Sensors have become important components of modern engineering systems, playing a central role in industrial automation, healthcare, environmental monitoring, transportation, smart cities, and consumer electronics [1], [2]. By converting physical quantities into digital or analog signals, sensors provide the essential data needed for monitoring, control, optimization, and decision-making in cyber-physical systems (CPS) [3], [4]. The rapid advancement of the Internet of Things (IoT) and Industry 4.0 paradigms has further accelerated the deployment of sensor networks at unprecedented scales, connecting billions of devices, and driving data-centric innovations across domains [5], [6].

However, as reliance on sensors increases, so does the vulnerability of these systems to faults and failures [7], [8]. Sensor faults can arise from various sources, including hardware aging, environmental conditions, communication failures, calibration drift, electromagnetic interference (EMI), or software errors [9], [10]. Even transient or intermittent faults can propagate through interconnected systems, compromising performance, causing false alarms, leading to financial losses, or most critically threatening human safety [9], [11].

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TABLE 1. Comparative Analysis of Recent SFD and Mitigation Reviews

Review attribute	Li et al. [28]	Trapani and Longo [8]	Chauhan et al. [10]	Jombo et al. [29]	Teh et al. [11]	This paper
Focus scope	SFD Advances	SFD Methods	SFD Breakthroughs	SFD Methods / Challenges	Sensor Data Quality	**Comprehensive SFD & Mitigation**
Fault detection	✓✓✓	✓✓✓	✓✓✓	✓✓✓	✓	✓✓✓✓
Fault isolation	✓✓	✓✓	✓✓	✓	–	✓✓✓✓
Fault estimation	✓	✓	✓	✓	–	✓✓✓✓
Fault types addressed	✓✓✓	✓✓✓	✓✓✓	✓✓	✓	✓✓✓✓
Analysis of causes	✓✓✓	✓✓	✓✓	✓	–	✓✓✓✓
Impact evaluation	✓✓	–	–	–	✓	✓✓✓✓
Mitigation approaches	+	–	–	–	–	✓✓✓✓
Technique comparison	✓✓	✓	✓✓	✓	–	✓✓✓✓
Sensor types covered	✓✓	✓✓	✓✓	✓	✓✓	✓✓✓✓
Domain generality	High	High	High	High	High	**Cross-Domain**
Reference breadth	142	95	158	22	106	**198**
Publication year	2020	2023	2024	2024	2020	2025

Legend: – = Not Addressed; + = Brief Mention; ✓= Addressed; ✓✓= Moderately Addressed; ✓✓✓= Well Addressed; ✓✓✓✓= Thoroughly Addressed. **SFD:** Sensor Fault Diagnosis. **Mitigation:** Compensation, Rectification, Recovery, Fault tolerance.

Real-world incidents clearly demonstrate the serious consequences of sensor faults. The CrowdStrike Falcon sensor outage, caused by a faulty software update, triggered widespread Windows system crashes, disrupting essential sectors, such as healthcare, aviation, and finance, with healthcare alone incurring $1.94 billion in direct losses [12], [13]. In the automotive domain, airbag sensor malfunctions, such as those involved in the Takata recall, caused at least 28 deaths and over 400 injuries due to unintended deployments or failures to deploy [14]. Aviation has also experienced catastrophic sensor-driven disasters; notably, the Boeing 737 MAX crashes in 2018 and 2019 were linked to faulty angle-of-attack sensors, triggering the Maneuvering Characteristics Augmentation System (MCAS) and resulting in the tragic loss of hundreds of lives [15]. In high-risk industrial environments, such as oil and gas, the Deepwater Horizon disaster in 2010 highlighted the dependence on accurate sensor data for critical operational decisions [16]. These events make it clear that sensor failures are not merely technical issues, they carry significant human, economic, and societal consequences.

To address these challenges, an extensive body of research has emerged, developing fault detection, diagnosis, and tolerance methods ranging from statistical techniques and physics-based modeling to advanced data-driven approaches, including artificial intelligence (AI) and deep learning (DL) [17], [18], [19]. Model-based strategies leverage residual generation, parity relations, and analytical redundancy to detect inconsistencies between measured and estimated system behaviors [20], [21], [22], while data-driven approaches extract fault signatures from historical data to detect anomalies without requiring detailed system models [8], [23], [24]. Recent developments have explored hybrid methods that combine physics-based and AI-based techniques, improving adaptability and robustness in complex and dynamic environments [10], [25], [26]. Despite this progress, current state-of-the-art technologies face persistent limitations. Data-driven methods require large, high-quality labeled datasets, which are often unavailable, especially for rare or safety-critical fault conditions [8], [10], [27]. Physics-based models depend on an accurate knowledge of system dynamics, which may be impractical in complex systems with uncertainties [7], [9], [17], [27].

Sensor-fault diagnosis (SFD) has become a key area of research. Recent surveys by Li et al. [28], Trapani and Longo [8], Chauhan et al. [10], and Jombo et al. [29] have advanced understanding of detection and diagnosis techniques (FDD). However, as summarized in Table 1, these studies primarily focus on the detection and isolation aspects of faults. In many cases, mitigation and compensation measures are only briefly mentioned as future work or generic "corrective actions" without detailed technical analysis. Furthermore, prior reviews often limit their scope to specific sensor types (e.g., air data sensors, industrial sensors) or narrow application domains, which reduces their generalizability for broader CPS and cross-domain applications.

In addition, critical aspects such as the systematic study of fault causes, impact analysis, and structured comparative evaluation of strengths and tradeoffs among existing methods are often overlooked or treated superficially. In many cases,

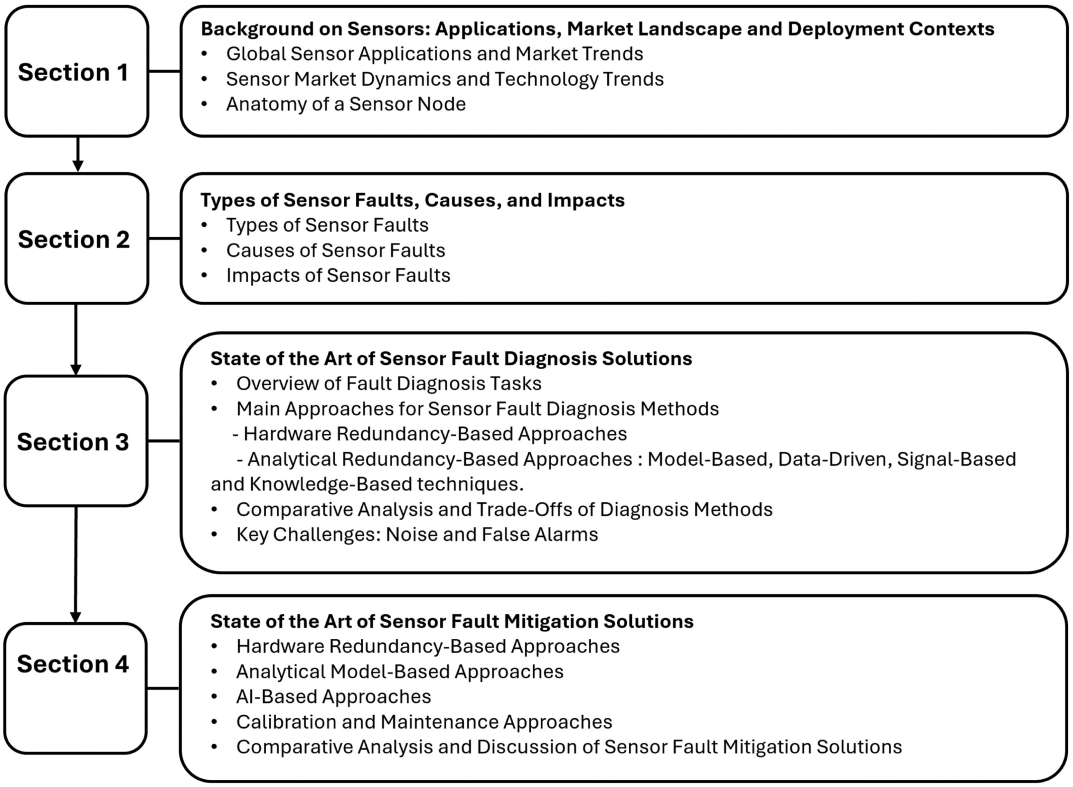


FIGURE 1. Structure and workflow of the review.

reviews provide descriptive lists of methods but lack a deeper discussion of practical challenges such as implementation cost, scalability, computational complexity, and integration with modern distributed or networked systems.

In response to these gaps, this article offers a comprehensive and structured review that addresses the full lifecycle of sensor fault management, from fault characterization and cause analysis to robust detection, precise diagnosis, and practical mitigation strategies. Unlike previous works, our review places equal emphasis on how to handle faults after detection, detailing solutions such as hardware redundancy, model-based compensation, AI-driven reconstruction, calibration, and maintenance techniques. Another key distinguishing feature is the broad scope: this review spans diverse sensor types and application domains, including automotive, aerospace, industrial automation, smart grids, environmental monitoring, and IoT networks, to present a generalizable framework that is relevant across disciplines. Furthermore, this work includes a clear comparative synthesis of existing approaches, highlights their practical tradeoffs, and identifies current limitations that must be addressed for real-world deployment.

This article presents a comprehensive and structured review of sensor faults and their management, with the aim of serving as a valuable reference for researchers and practitioners working toward the development of the next generation of resilient, adaptive, and trustworthy smart sensor systems. The rest of this article is organized as follows. Section II provides background on sensor applications, deployment contexts, market

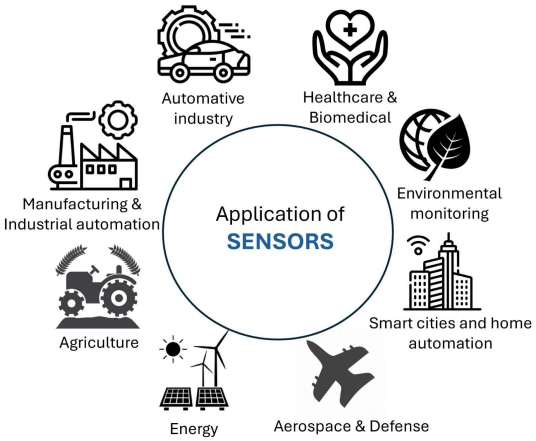


FIGURE 2. Illustrative overview of major application domains for modern sensor technologies.

trends, and fundamental sensing principles. Section III categorizes the main types of sensor faults, analyzes their root causes, and discusses their impact on system performance, reliability, and data integrity. Section IV reviews and compares the state-of-the-art techniques for sensor fault detection and diagnosis, highlighting key methodologies, strengths, and limitations. Section V focuses on fault mitigation strategies, examining both hardware-based and analytical approaches to restoring sensor functionality. Finally, Section VI concludes this article.

Fig. 1 provides a visual overview of the structure and logical flow of the article.

II. BACKGROUND ON SENSORS: APPLICATIONS, MARKET LANDSCAPE, AND DEPLOYMENT CONTEXTS

Sensors are the foundational interface between the physical world and digital systems. They act as transducers that detect, measure, and convert various physical, chemical, or biological stimuli, such as temperature, motion, light, pressure, and biochemical reactions, into interpretable electrical or optical signals [2], [30]. This translation is essential for monitoring, automation, and intelligent decision-making in nearly all sectors of modern technology. From sensors embedded in personal wearables to those integrated in critical infrastructure, their ubiquity and functionality underscore their indispensable role in enabling the IoT, Industry 4.0, healthcare, smart cities, and autonomous systems [1], [5], [31].

A. GLOBAL SENSOR APPLICATIONS AND MARKET TRENDS

The global integration of sensors has reached unprecedented levels, with applications spanning healthcare, manufacturing, transportation, agriculture, environmental monitoring, energy, aerospace, and defense (see Fig. 2). In *industrial automation*, sensors monitor parameters such as vibration, pressure, flow, and position, enabling predictive maintenance and real-time process control [5], [32]. In the *automotive sector*, an array of mechanical, thermal, optical, and MEMS-based sensors allows engine control, emissions reduction, safety features such as airbag deployment, and advanced driver-assistance systems such as light detection and range, radar, and ultrasonic systems [33], [34].

In *healthcare*, biosensors and wearable physiological sensors are now essential for diagnostics, drug delivery, and remote monitoring of chronic diseases [31], [35]. Applications range from glucose monitors to wearable Electrocardiogram (ECG) sensors and smart prosthetics. Meanwhile, the *environmental sector* utilizes electrochemical, optical, and biosensors to detect pollutants, monitor air and water quality, and track climate parameters [36], [37], [38]. In *smart cities*, sensors manage traffic flow, optimize energy use, monitor noise levels, and provide early warnings for hazards [39], [40].

In *precision agriculture*, smart sensors help optimize irrigation, monitor soil nutrient content, and predict pest infestations using drones and IoT systems [41], [42]. Similarly, in the *aerospace and defense industries*, sensors support structural health monitoring, navigation, target tracking, and threat detection in extreme environments [43], [44], [45]. Lastly, the rise of *Energy Systems 4.0* digitalization in energy generation and management is based on distributed sensor networks for grid stability, load forecasting, and integration of renewable sources [46], [47].

B. SENSOR MARKET DYNAMICS AND TECHNOLOGY TRENDS

The sensor market is growing rapidly. According to fortune business insights, the global sensor market was valued at approximately *USD 241.06 billion in 2024* and is projected to reach over *USD 450 billion by 2032*, driven by increased automation, electrification, and demand for real-time data [48].

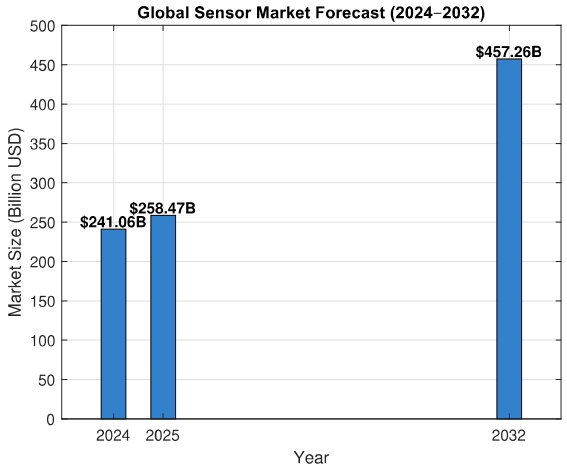


FIGURE 3. Global sensor market size and forecast from 2024 to 2032 (Conceptual visualization based on [48]).

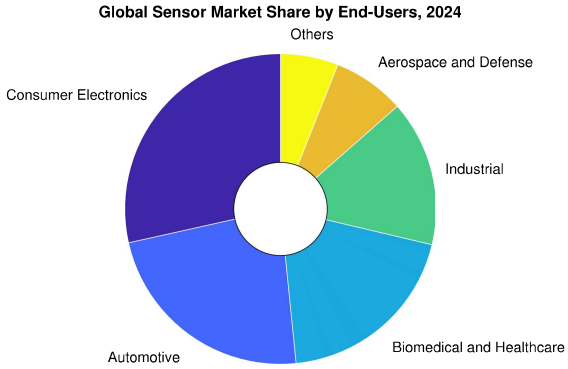


FIGURE 4. Global sensor market share by end-user application [48].

Figs. 3 and 4 conceptually illustrate the current market trajectory by size and end-user demand.

Key trends influencing this growth include the following.

1) *By sensor type:* Core sensor classes such as pressure, temperature, and position sensors remain dominant in industrial and automotive settings. However, rapid growth is observed in image sensors (for autonomous vehicles and augmented reality(AR)/virtual reality (VR)), biosensors (for diagnostics), and gas sensors (for environmental safety) [2], [49].
2) *By technology:* Innovations in microelectromechanical-systems (MEMS) and complementary metal-oxide semiconductor (CMOS) technologies have enabled the miniaturization and cost-efficiency of sensors while maintaining accuracy and reliability [33]. Furthermore, the integration of AI and edge computing has led to the emergence of smart sensors that can preprocess data locally, reducing latency and bandwidth usage [5].
3) *By end-user:* The consumer electronics segment is growing rapidly due to smart wearables and IoT devices, while the automotive and industrial sectors lead in terms of volume and investment. Healthcare remains a strategic growth area, particularly for remote and personalized medicine [31], [32].

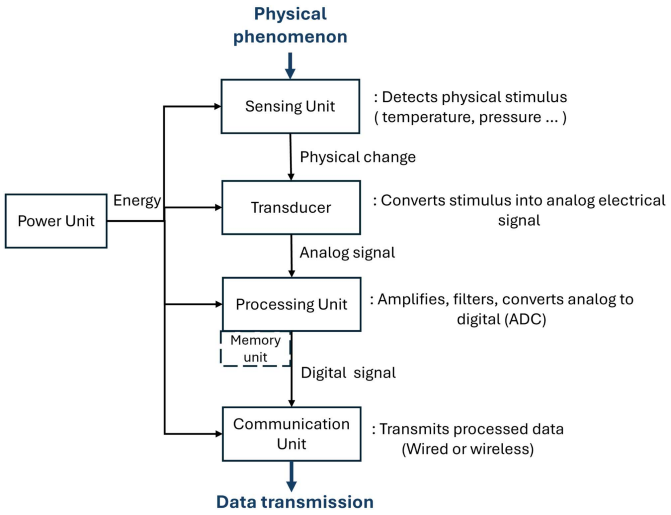


FIGURE 5. Generalized structure of a sensor node, showing core functional units and signal flow.

C. ANATOMY OF A SENSOR NODE

A sensor node is a self-contained unit designed to acquire data from its environment and communicate that data for processing or control. Although sensor configurations vary across domains, from industrial MEMS sensors to biomedical biosensors, their architecture typically includes a set of core subsystems that enable sensing, processing, communication, and power management [2], [5], [30].

Understanding the internal structure of sensor nodes is fundamental to evaluating reliability and diagnosing faults. Fig. 5 shows the generalized architecture of a sensor node, which typically comprises the following functional components.

1) *Sensing unit:* This is the primary interface between the sensor and its environment. It contains the physical or chemical sensing element that interacts with the target stimulus, such as temperature, pressure, light, gas concentration, and acceleration, and exhibits a change in a measurable property [36], [37]. For example, thermistors vary resistance with temperature, while piezoelectric materials generate voltage under mechanical stress.
2) *Transducer (signal conversion element):* The transducer converts the change detected by the sensing unit into an electrical signal. Depending on the sensor type, this might involve changes in resistance, current, voltage, or capacitance [1], [50]. For instance, photodiodes translate light intensity into current, and ion-selective electrodes convert ion concentration into potential.
3) *Signal conditioning and processing unit:* This unit prepares the raw signal for transmission and/or further computation. It typically includes analog signal conditioning elements, such as amplification, filtering, and impedance matching, followed by an analog-to-digital converter (ADC) [2]. In smart or embedded sensors, this module may include a microcontroller or digital signal processor capable of executing data compression, noise filtering, feature extraction, or embedded machine learning (ML) for local decision-making [35].
4) *Communication unit (transceiver):* The processed signal must be transmitted to an external system, such as a data acquisition server and control loop. This is enabled by a transceiver, which supports wired (for example, I2C, SPI, CAN, Ethernet) or wireless (for example, Wi-Fi, Zigbee, LoRa, Bluetooth) communication protocols [2], [39]. Wireless interfaces are especially important in distributed sensing environments, such as IoT, wearable devices, and environmental monitoring systems.
5) *Power supply unit:* All sensor operations require a power source. Power may be supplied by batteries, direct current (DC) from power grids, or energy harvesting techniques such as photovoltaic cells and vibrational energy scavengers [40], [51]. In remote or embedded scenarios, power efficiency becomes critical, necessitating ultra-low power designs, and intelligent sleep cycles.
6) *Memory unit (optional):* Many sensor nodes integrate volatile or nonvolatile memory components for temporary data storage, buffering, logging, or storing firmware and calibration parameters [1], [2]. Flash memory is often used for program storage, while RAM may support temporary data handling.

III. TYPES OF SENSOR FAULTS, CAUSES, AND IMPACTS

Sensors despite their increasing sophistication are susceptible to various types of faults that can compromise the accuracy and reliability of their measurements [7]. Understanding these types of fault, their origins, and their consequences is crucial to developing effective diagnostic and mitigation strategies [8].

A. TYPES OF SENSOR FAULTS

Among the most reported sensor faults is the *bias fault*, which causes a consistent offset in the sensor output across its operating range. Such faults can originate from initial calibration errors, long-term thermal effects, or progressive aging of sensor components [52], [53], [54], [55]. For instance, Zhang et al. [54] demonstrated thermal bias drift in MEMS accelerometers and proposed mitigation methods. Undetected bias faults can accumulate in control loops, degrading performance, or compromising safety in critical applications [56]. Table 2 presents both graphical and mathematical representations of a bias fault, where the faulty output $y_f(t)$ deviates by a constant offset from the true value $y_h(t)$ after the fault occurs at t_f.

A *drift fault* is similar but unfolds gradually: the sensor output slowly changes under constant conditions due to material fatigue, environmental stressors, or component degradation [57], [58], [59]. Drift is especially challenging in long-term missions where recalibration is impractical, as in satellite systems [57]. It can also mislead data-driven models, which may misinterpret drift as a real trend [60], [61]. Zhao et al. [58] proposed an improved ensemble method combining

TABLE 2. Bias Fault: Graphical and Mathematical Representation

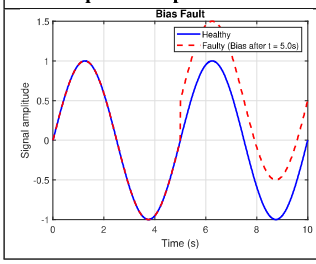
Graphical Representation	Mathematical Representation
	$y_f(t) = y_h(t), \quad t < t_f,$ $y_f(t) = y_h(t) + b, \quad t \geq t_f,$ b : constant bias.

TABLE 3. Drift Fault: Graphical and Mathematical Representation

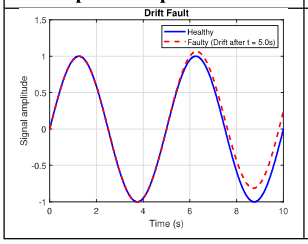
Graphical Representation	Mathematical Representation
	$y_f(t) = y_h(t), \quad t < t_f,$ $y_f(t) = y_h(t) + d(t - t_f), \quad t \geq t_f$ d : drift rate (constant).

TABLE 4. Loss of Accuracy: Graphical and Mathematical Representation

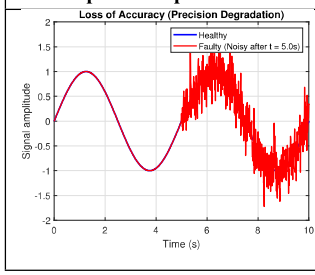
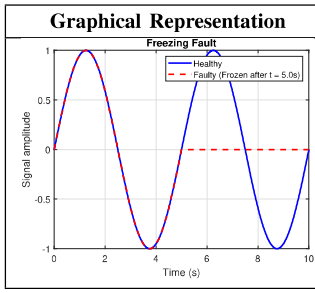
Graphical Representation	Mathematical Representation
	$y_f(t) = y_h(t), \quad t < t_f,$ $y_f(t) = y_h(t) + d_i(t), \quad t \geq t_f,$ $\lvert d_i(t)\rvert \leq \bar{d}_i, \quad \dot{d}_i(t) \to 0,$ $\bar{d}_i \in [-\epsilon_{\min}, \epsilon_{\max}],$ $\epsilon_{\min}, \epsilon_{\max} > 0.$

TABLE 5. Freezing Fault: Graphical and Mathematical Representation

Graphical Representation	Mathematical Representation
	$y_f(t) = y_h(t), \quad t < t_f,$ $y_f(t) = y_h(t_f), \quad t \geq t_f.$

long-short-term memory (LSTM) networks and support vector machines (SVMs) to compensate for drift in gas sensors, achieving superior classification performance over multiyear datasets. Similarly, Saeed et al. [60] and Jan et al. [61] developed ML frameworks for real-time detection of sensor drift faults, demonstrating the feasibility of using embedded systems like Raspberry Pi for online monitoring. In Table 3, both graphical and mathematical representations of a drift fault are presented. The fault is characterized by a drift rate d, which causes the faulty signal to deviate linearly from the healthy signal over time after the onset of the fault.

Loss of accuracy, also known as precision degradation, is characterized by an increase in the variability or uncertainty of the measurements of a sensor, even when the average output remains close to the true value [62], [63]. This decline in measurement quality can be caused by elevated noise levels within sensor electronics, physical wear and tear affecting the sensor's sensitivity, or external environmental disturbances that introduce random fluctuations into the signal [64], [65]. The consequences of reduced accuracy are substantial: they diminish confidence in individual measurements and make it increasingly difficult to detect small but meaningful changes or subtle patterns in collected data [63], [65]. For example, Ma et al. [65] highlighted the difficulty of distinguishing true structural damage from sensor performance degradation in civil infrastructure, emphasizing the importance of isolating degraded sensors to prevent false alarms. Table 4 presents both graphical and mathematical representations of a loss of accuracy fault. Specifically, the fault introduces a bounded degradation in the accuracy of the sensor signal, where the error $d_i(t)$ remains within a predefined range $\bar{d}_i$, and the error rate approaches zero after the fault onset. This characterization is essential for designing robust monitoring systems that can maintain performance despite increased measurement uncertainty.

Freezing, also known as a stuck fault, refers to a condition in which a sensor persistently produces a constant or fixed value, regardless of changes in the underlying physical quantity it is intended to measure [29], [66], [67]. This fault effectively "freezes" the sensor signal, making it insensitive to the variations and dynamics of the real world. Freezing faults can arise from several underlying mechanisms, including hardware malfunctions such as ADC failures, memory register corruption, and signal pathway interruptions within sensor electronics [66], [67]. The importance of addressing freezing faults has been emphasized in various domains. For example, Ni et al. [68] provided a taxonomy of sensor network data faults, identifying freezing as one of the most common and disruptive types. Li and Yang [67] developed detection algorithms specifically for linear stochastic systems affected by stuck sensor faults, highlighting the statistical challenges posed by constant value signals in dynamic environments. In Table 5, both the graphical and mathematical representations of a freezing fault are presented. Specifically, the sensor output freezes at its value at the start of the fault, remaining constant for all time after t_f, such that $y_f(t) = y_h(t_f)$ for all $t \geq t_f$. This behavior necessitates specialized detection approaches that can differentiate between true constant physical phenomena and faulty sensor outputs.

Intermittent faults are a type of sensor fault that occurs irregularly and without a clear pattern, separated by periods of normal operation [69], [70], [71]. These faults are difficult to detect because they appear only briefly and unpredictably. Common causes include loose electrical connections that temporarily interrupt signals, short-lived environmental

TABLE 6. Intermittent Fault: Graphical and Mathematical Representation

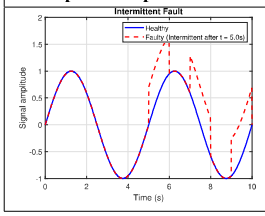
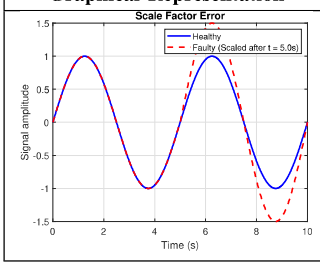
Graphical Representation	Mathematical Representation
	$y_f(t) = y_h(t), \quad t < t_f,$ $y_f(t) = \begin{cases} y_h(t) + i, & (t - t_f) \bmod T < \tau, \\ y_h(t), & \text{otherwise} \end{cases}$ $t \geq t_f,$

TABLE 7. Scale Factor Error: Graphical and Mathematical Representation

Graphical Representation	Mathematical Representation
	$y_f(t) = y_h(t), \quad t < t_f,$ $y_f(t) = s \cdot y_h(t), \quad t \geq t_f,$ $s:$ scale factor,

disturbances (such as EMI or mechanical vibrations), or components that fail only under specific conditions [7], [72]. Because these faults do not appear consistently, they often go unnoticed during standard tests. As a result, they can cause unreliable sensor data, erratic system behavior, and incorrect measurement interpretation in real world applications [73], [74]. In Table 6, both the graphical and mathematical representations of an intermittent fault are presented. Specifically, this fault mode is characterized by periodic occurrences of a fault amplitude i during an active duration τ within each fault period T, alternating with periods in which the sensor behaves normally. Mathematically, this can be expressed as a time-dependent function in which the faulty signal switches between the healthy state and the fault state over repeated cycles, posing challenges for timely and reliable detection.

Finally, *scale factor error* represents a deviation in the linear relationship between the change in the measured quantity and the corresponding change in the sensor output. In this fault mode, the sensor sensitivity deviates from its specified value, leading to a consistent over- or underestimation throughout the measurement range. Specifically, the sensor output is multiplied by a constant scale factor s, where $s > 1$ amplifies the signal and $0 < s < 1$ attenuates it. This error results in proportional inaccuracies that increase with the magnitude of the measured value, which is particularly critical for applications operating over wide dynamic ranges. The causes of scale factor error are diverse and include manufacturing imperfections in the transduction mechanism, gradual aging and drift of internal components, environmental influences such as temperature and humidity, and improper calibration procedures [75], [76], [77]. For example, Guo and Zhong [75] addressed calibration and compensation strategies for scale factor errors in dynamically tuned gyroscope positioning systems, while Zheng et al. [76] proposed modeling and compensation techniques for inertial sensor errors. In precision measurement applications, scale factor nonlinearities can limit system accuracy, for instance, Shi et al. [78] analyzed uncertainty propagation in roundness measurements, underscoring how scale errors interact with harmonic suppression problems.

In Table 7, both the graphical and mathematical representations of the scale factor error are presented. After the onset of the fault, the sensor output is described as

$$y_f(t) = s \cdot y_h(t)$$

where $y_h(t)$ is the true physical signal and $y_f(t)$ is the measured output.

B. CAUSES OF SENSOR FAULTS

Sensor faults can come from many different sources, including environmental factors, wear and tear, manufacturing flaws, improper use, and security weaknesses [8], [10]. Each of these areas covers a variety of specific causes that can lead to the different types of faults mentioned earlier.

Environmental factors are among the most prevalent contributors to sensor faults. Variations in ambient temperature can shift material properties or electronic behavior, leading to bias drifts, sensitivity loss, or scale factor errors [79], [80]. For instance, in MEMS resonators, combined temperature and humidity changes can significantly alter resonance frequencies and damping, degrading measurement precision [80]. Humidity also plays a critical role by inducing corrosion, swelling polymers, or changing electrical conductance within circuits. Tong et al. [81] demonstrated how humidity fluctuations affect the performance of piezoelectric micromachined ultrasonic transducers (PMUTs), shifting resonance and acoustic behavior. In high-radiation environments, such as aerospace or nuclear facilities, cumulative exposure can damage semiconductor junctions or piezoelectric films, causing irreversible performance loss. EMI from nearby electronics or power systems can inject noise and distort sensor signals [82]. Mechanical shocks and continuous vibrations can loosen connections or misalign internal components, especially in inertial sensors such as gyroscopes and accelerometers [75], [76]. For example, Rengarajan and Schott [79] quantified how solar angle variability can cause up to 40% uncertainty in satellite remote sensing indices, stressing the need for robust compensation and calibration routines.

Wear and tear is another common cause of sensor faults, particularly in sensors with mechanical components or those that rely on chemical reactions [83], [84]. Mechanical sensors, such as MEMS devices, are particularly susceptible to fatigue and material degradation when subjected to repeated cycles of operation, which can manifest as calibration, drift, bias errors, and ultimately loss of functionality [84]. For example, cyclic stresses on MEMS structures can induce crack formation, stiction, or material fatigue, reducing their long-term reliability [84]. Chemical sensors, including metal oxide

gas sensors and catalytic sensors, also experience performance degradation over time due to the depletion of active sensing materials, contamination from environmental exposure, or accumulation of defects in the sensing layer [85]. Dennler et al. [86] highlighted how drift in metal oxide gas sensors over prolonged use can confound gas classification tasks, underscoring the importance of accounting for temporal degradation in both sensor design and data analysis. Even purely electronic components within sensors are not immune to aging; effects such as electromigration, dielectric breakdown, and contact degradation can alter the characteristics of the sensor response over time, leading to gradual drift in readings [75], [76]. Wear mechanisms extend beyond the sensing element to auxiliary components such as bearings in rotary sensors or motors, where lubrication degradation and tribological wear can trigger secondary faults [83]. Lu et al. [83] emphasized the importance of early wear detection in mechanical systems, proposing microsensing approaches to monitor tribological phenomena for improved condition monitoring. Furthermore, specific applications, such as machining operations, require targeted wear detection systems. Rakkiyannan et al. [87] developed a thick film silver-polyester sensor system for real-time tool wear detection, illustrating the potential of integrated sensing technologies to detect wear at the source and prevent downstream failures.

Manufacturing defects can also be a significant source of sensor faults. Imperfections introduced during fabrication, such as impurities in materials, microstructural defects, and assembly defects, can result in initial biases, reduced accuracy, or even premature sensor failure [88], [89]. Variability in material properties or inconsistencies in manufacturing processes can cause substantial performance differences between individual sensors of the same model, complicating both calibration and system integration [90], [91]. Rigorous quality control and certification processes are, therefore, essential to minimize manufacturing-induced faults. Martins et al. [89] reviewed global calibration and certification practices for industrial sensors, emphasizing their importance for consistent performance. Early detection of defects, through incoming inspection or automated in-line testing, can prevent costly downstream failures and improve the overall reliability of deployed systems [8], [91]. As manufacturing systems become more complex, their susceptibility to introducing hidden faults increases, further underscoring the need for robust monitoring and validation throughout the production pipeline [88].

Operational misuse represents a category of causes related to the way sensors are deployed, handled, and maintained in the field [90], [91]. Exceeding specified operational limits, such as temperature, pressure, mechanical load, or electrical input, can cause irreversible damage or long-term degradation of sensor performance [28], [92]. Improper installation, such as misalignment, poor grounding, and incorrect mounting, can introduce mechanical or electrical stresses that degrade accuracy or shorten useful life [89]. In addition, failure to perform regular calibration and maintenance can exacerbate minor drift or bias, eventually leading to critical failures [8], [89]. The importance of proper training for personnel handling sensor systems has been emphasized in several domains, particularly in industries such as construction, manufacturing, and electric motor drives [88], [90]. Wang et al. [90] identified the lack of standardized practices and advanced training as key barriers to sensor adoption and reliable operation in the construction sector. Many operational faults are preventable through careful adherence to manufacturer guidelines, regular inspection, and the use of automated diagnostic tools to monitor sensor health [8].

Security vulnerabilities have become a major source of sensor faults due to the growing integration of sensors into connected environments such as the IoT and CPS [9], [93]. Unlike faults caused by physical wear or environmental stress, security threats intentionally target the functionality and integrity of sensor systems. These attacks aim to generate false measurements, disrupt system behavior, or extract sensitive information. Attackers can exploit weaknesses in hardware, software, or communication protocols using methods such as spoofing, false data injection (FDI), signal jamming, and firmware manipulation, leading to apparent sensor faults that degrade the reliability of the system [9], [93], [94]. Bindra and Aggarwal [93] discussed how increased autonomy and interconnectivity of CPS systems introduce new security risks, emphasizing the need for integrated solutions that combine detection, control, and security. In IoT networks, Litoussi et al. [9] reviewed cryptographic techniques and system-level protections to ensure data confidentiality, integrity, and availability, the key to preventing critical system failures. An especially dangerous form of attack is the FDI attack, widely studied in smart grids and power systems. Husnoo et al. [95] surveyed recent developments in this area, showing how altered sensor data can mislead control decisions and disrupt grid operations. Shahid et al. [96] introduced detection and mitigation techniques using dynamic "dummy value" models, successfully neutralizing even sophisticated FDI attacks in IEEE benchmark test systems. Sensor security challenges also extend to autonomous vehicles. Islam et al. [97] reviewed threats to automotive sensing systems, including GPS spoofing, CAN bus attacks, and adversarial ML, highlighting the serious risks these pose to vehicle control and safety. In industrial applications, Rathee et al. [98] proposed a blockchain-based framework to secure IoT sensor communications in Industry 4.0, demonstrating the value of decentralized technologies in building trust in sensor networks. The impact of cyberattacks on sensor systems goes beyond corrupting measurements; it can cause large-scale operational failures and erode user trust. As recent research shows, securing sensor systems is now a critical part of fault prevention, detection, and recovery in CPS and IoT systems [8], [93]. Without strong safeguards, security breaches can mimic or worsen other fault types such as bias, drift, freezing, and intermittent failures, making them harder to detect and mitigate.

C. IMPACTS OF SENSOR FAULTS

The *impacts* of sensor faults can be significant and varies greatly depending on the application, the nature of the fault, and whether fault-tolerant mechanisms or redundancies are in place [99], [100]. Fundamentally, sensor faults degrade the accuracy and reliability of collected data, which can then propagate through control loops, decision-making systems, and broader networks, often resulting in suboptimal or even dangerous outcomes [100], [101]. These effects are particularly critical in mission-critical or safety-critical applications, where faulty data can trigger cascading failures or unsafe behaviors.

In healthcare, sensor faults can be life-threatening. Hong et al. [102] analyzed the impact of sensor failures in smart homes and wireless body area networks, showing how such failures compromise the trustworthiness of activity recognition and assistive systems for the elderly. Similarly, Sathasivam et al. [103] highlighted challenges in healthcare sensor systems, noting that faults can compromise diagnostics, continuous monitoring, and treatment delivery, underscoring the need for robust sensor networks.

In the energy sector, Im et al. [100] demonstrated that incipient sensor faults in commercial building energy systems can result in severe inefficiencies. Their large-scale simulations showed deviations in heating energy up to 314% and deviations in cooling energy up to 65% compared to baseline, highlighting the significant implications for cost and carbon footprint. Zhang et al. [99] further emphasized that poor sensor design, placement, and calibration can undermine fault detection and diagnosis (FDD) in buildings, limiting the effectiveness of energy-saving technologies.

For power systems, Guo et al. [101] modeled cyber-physical power grids and demonstrated how faults or attacks on sensor networks can propagate throughout the system, leading to control failures and widespread outages. Mehmood et al. [104] examined power electronics inverters and showed that sensor faults significantly degrade power quality, highlighting the importance of fault modeling in electrical reliability engineering.

In the aerospace domain, Wang et al. [105] proposed a spatiotemporal information fusion model to enhance the reliability of spacecraft sensor data. Their method, validated on satellite flywheel data, demonstrated accuracy in anomaly detection 93% and extended mean failure time, underscoring the operational and economic risks associated with unreliable sensor data in space missions.

Transportation networks are also vulnerable to sensor faults. Salari et al. [106] developed a novel optimization model to account for time-dependent sensor failures when locating traffic counting sensors for origin-destination estimation. They showed that ignoring sensor failure leads to substantial information loss and poor traffic management decisions, while robust sensor placement strategies improve data reliability and system resilience.

In environmental monitoring, Khizar et al. [37] emphasized that sensor faults can result in false alarms or missed detections, compromising pollution control, and public health measures. Given the simplicity and cost effectiveness of environmental sensors, their reliability is often taken for granted, making rigorous calibration and maintenance essential.

CPS and IoT networks are particularly susceptible to the propagation of sensor faults. Castaño et al. [88] reviewed sensor reliability challenges in IoT-based CPS, highlighting how even localized sensor faults can escalate into system-wide failures if not promptly detected and mitigated. This is especially critical in applications such as smart grids, autonomous vehicles, and manufacturing lines.

In general, sensor faults not only affect the accuracy of local measurements, but also introduce systemic risks, leading to degraded performance, economic losses, safety hazards, and reputational damage. Effective fault management, including detection, isolation, and compensation, therefore, becomes an indispensable requirement across industries, particularly in high-assurance sectors where human safety, critical infrastructure, or environmental sustainability are at stake. Table 8 presents a comprehensive overview of common sensor fault types, summarizing their definitions, root causes, affected subsystems, frequency of occurrence, sensitive applications, and representative studies. This classification helps to understand the manifestation of faults in various domains and guides the development of tailored detection and mitigation strategies.

IV. STATE OF THE ART OF SFD SOLUTIONS

The increasing complexity and criticality of modern systems that rely on sensor data require robust methodologies to identify and manage sensor faults. SFD is a multifaceted process designed to ensure the accuracy, reliability, and continuity of sensor-based measurements. This process encompasses several interrelated tasks that range from initial detection of anomalies to precise estimation and classification of faults [8], [10], [28]. As sensors operate in increasingly harsh and dynamic environments, especially within industrial, agricultural, and autonomous systems, the ability to detect, isolate, and compensate for sensor faults is essential to maintain overall system integrity and operational safety [1].

Various methods have been developed to perform SFD, typically classified according to their underlying principles and the nature of data they utilize. At a high level, these methods are categorized into hardware redundancy techniques and analytical redundancy approaches, the latter being further subdivided into model-based, data-driven, signal-based, and knowledge-based strategies. Before detailing these classes, it is useful to delineate the key tasks that comprise the diagnostic process.

A. OVERVIEW OF FAULT DIAGNOSIS TASKS

An effective SFD involves several fundamental tasks, each providing an increasing depth of information about the health of the system. These tasks, illustrated in Fig. 6, are generally performed in sequence, although some methods can address multiple tasks simultaneously [8], [111].

TABLE 8. Summary of Common Sensor Fault Types, Causes, and Characteristics

Fault Type	Description	Primary Causes	Affected Unit(s)	Frequency	Vulnerable Sensors/Applications	References
Bias Fault	Constant offset from true value	• Initial calibration error • Aging • Thermal/ environmental effects	Sensing unit, transducer, processing unit	Common	• MEMS sensors • Temp/humidity • Battery systems • HVAC	[54] [53] [107]
Drift Fault	Gradual deviation over time	• Environmental stress • Material fatigue • Component aging	Sensing unit, transducer	Frequent	• Gas/temp sensors • UAVs, satellites • IoT devices	[57] [61] [59]
Loss of Accuracy	Increased variability and Precision Loss	• EMI • Physical wear • Clogged elements	Sensing, processing, conditioning units	Common	• Flow/pressure sensors • Infrastructure • Data centers	[64] [65] [63]
Freezing Fault	Output stuck at fixed value	• Signal path failure • Comms/power glitch • ADC/memory fault	Sensor, processing, communication units	Common	• Satellite/robotic • WSNs • Chemical gauges	[66] [108] [67]
Intermittent Fault	Irregular fault occurrence	• Loose wiring • EMI/vibration • Power spikes	Power, comms, processing units	Common/ Unpredictable	• WSNs • Motor drives • IoT sensors	[74] [109]
Scale Factor Error	Proportional measurement error	• Calibration issues • Temp/humidity drift • Shock/vibration	Sensor, transducer, processing unit	Comman	• Gyros • Inertial nav • Accelerometers	[110] [76] [77]

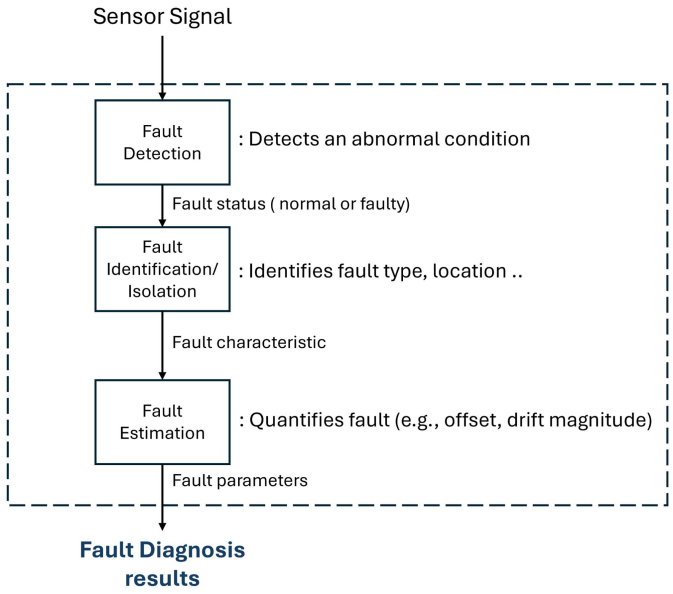


FIGURE 6. Fault diagnosis process in sensor systems, comprising detection, isolation, and estimation tasks.

1) *Fault detection:* The primary objective of this task is to determine whether a fault has occurred. This typically involves analyzing sensor output to identify deviations from expected behavior, using methods such as thresholding, statistical monitoring, and residual evaluation against a reference model [8], [10]. The output is typically binary, indicating the presence or absence of a fault.
2) *Fault isolation/identification:* After detecting a fault, the next step is to determine its origin. This step seeks to identify the specific sensor (or group of sensors) or the monitored system that is malfunctioning. In many cases, this also includes identifying the type of fault, such as bias, drift, and stuck conditions [28], [111].
3) *Fault estimation:* This task focuses on quantifying the characteristics of the fault, such as its magnitude, duration, and dynamics. For example, estimating a sensor's bias or drift rate enables corrective actions such as signal reconstruction and control reallocation. Estimation is often achieved through advanced observers, regression methods, or learning-based approaches [28], [111].

It is important to note that different diagnostic techniques vary in their ability to perform these tasks. Basic techniques may be limited to fault detection, while more sophisticated methods, such as Kalman filters and deep neural networks, can simultaneously detect, isolate, and estimate faults [10], [111]. The following sections classify and review these approaches based on their methodological foundations and application capabilities.

B. MAIN APPROACHES FOR SFD

SFD methods can be broadly categorized based on their fundamental principles and the type of data they use. This classification provides a systematic framework to analyze the advantages, limitations, and application contexts of each method. At a high level, these methods are divided into two primary categories: *hardware redundancy* methods and *analytical redundancy* methods. The former relies on physical

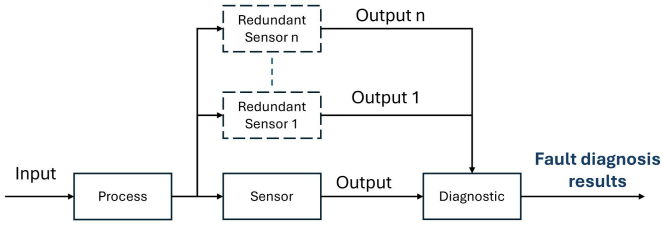


FIGURE 7. Concept of hardware redundancy for SFD. Multiple sensors measuring the same variable enable cross-validation to diagnose faulty units.

duplication of sensing elements, while the latter uses computational models or algorithms to infer sensor health. Each category offers unique benefits and tradeoffs in terms of fault detection, isolation, and estimation capabilities [10], [28], [112].

1) HARDWARE REDUNDANCY-BASED APPROACHES

Hardware redundancy is a widely used approach that involves deploying multiple physical sensors to monitor the same variable or process parameter, as illustrated in Fig. 7. This method is grounded in the principle that discrepancies among redundant sensor outputs can be exploited to detect and isolate faults. For example, if three temperature sensors are used to monitor the same point, a significant deviation in the output of one sensor relative to the others may indicate a fault [111].

Fault detection in this context is achieved by continuously comparing sensor outputs and identifying anomalies through techniques such as thresholding, pairwise comparison, and residual generation. Fault isolation is often achieved using voting schemes (e.g., majority voting and median filters) or deviation analysis relative to consensus values [112]. These approaches are effective in identifying the faulty unit, especially when faults are uncorrelated and sensor failures are rare.

Although hardware redundancy offers high diagnostic reliability and real-time responsiveness, it incurs significant costs associated with duplication of sensing components, wiring, calibration, and maintenance [8], [10]. Moreover, it may be infeasible in applications with strict restrictions on weight, space, and power, such as aerospace or mobile robotics [111].

Fault estimation using hardware redundancy is less direct. Typically, the faulty reading is discarded and its value is substituted using interpolation, averaging, or model-based estimation derived from the remaining healthy sensors. Thus, while effective for detection and isolation, hardware redundancy is less suitable for precise fault quantification unless combined with analytical techniques.

Despite these limitations, hardware redundancy remains a cornerstone in safety-critical domains where sensor failure could have catastrophic consequences, such as in nuclear power plants, space, and medical devices [10], [28], [62].

2) ANALYTICAL REDUNDANCY-BASED APPROACHES

Analytical redundancy methods, often referred to as software-based fault diagnosis, use mathematical models or data-driven techniques to infer sensor behavior without relying on physical sensor duplication. These methods take advantage of known relationships among sensor readings, control inputs, and system outputs under normal operating conditions to generate expected sensor values. Discrepancies between expected and actual values, referred to as residuals, signal the presence of faults [8], [10].

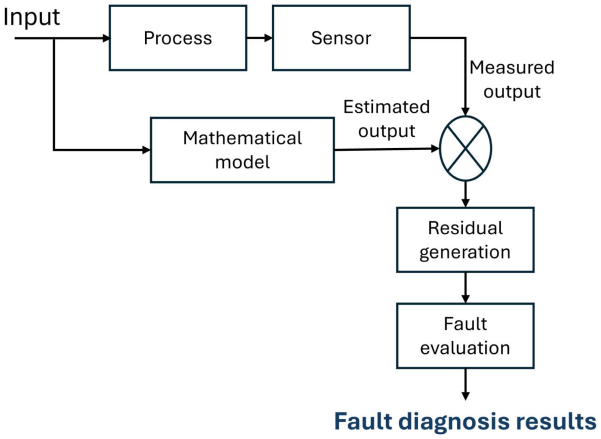


FIGURE 8. Structure of model-based for SFD.

Compared to hardware redundancy, analytical methods are generally more cost-effective, scalable, and adaptable to various systems. However, their effectiveness heavily depends on the accuracy and robustness of the model to uncertainties and disturbances [28].

a) Model-based methods: Model-based techniques are a key category of analytical redundancy approaches. They rely on physical or mathematical models derived either from first principles or from system identification techniques. These models predict system behavior, typically system outputs or internal states, which are then compared with actual sensor measurements to generate residuals used for fault detection and diagnosis [113], [114]. The general structure of model-based fault diagnosis is illustrated in Fig. 8, which outlines the flow from system inputs and models to residual generation and fault evaluation. This visual representation helps to clarify the sequential steps involved, including model-based prediction, residual computation, and decision logic for identifying faults.

Observers: Observer-based methods utilize state observers, such as Luenberger observers, Kalman filters, and sliding-mode observers (SMOs), to estimate sensor outputs or internal states. The difference between the estimated and actual measurements forms the residual used for fault detection and estimation [115], [116], [117]. For example, Huang et al. [114] applied an adaptive sliding-mode observer (ASMO) to identify current sensor faults in dc charging piles, effectively distinguishing between incipient, significant and complete failures. Robust observer design is a growing focus, aiming to improve diagnostic reliability under model uncertainties and disturbances. Yu et al. [118] and Wang et al. [119] reviewed the importance of robust observer structures in battery systems and sensor networks, respectively.

Parity equations: Parity space methods involve deriving algebraic or differential consistency checks that must hold true under fault-free operation. When these conditions are violated, the resulting residuals indicate a potential fault [70], [109], [120]. Lee [121] demonstrated the use of parity equations in a motorized power steering system, showing high fault detectability and isolability depending on the sensor configuration. Hua and Hongyue [122] proposed a decoupled-parity equation approach combined with parameter estimation, effectively isolating faults in aerospace applications. These methods offer strong theoretical underpinnings but can be sensitive to noise and modeling errors.

The efficacy of model-based methods has been demonstrated in numerous applications. Yu et al. [118] used a recursive least squares estimator and an unscented Kalman filter (UKF) to detect voltage and current sensor faults in electric vehicle battery systems, achieving robust results in simulation and experimentation. Similarly, Sravani et al. [123] implemented a linear parameter varying observer to detect multiple types of fault in flowmeter sensors, with fast and accurate detection in varying disturbances. Hybridization is an emerging trend in model-based diagnosis. Guc and Chen [124] proposed a physics-informed transfer learning approach that combines dynamic mode decomposition with control and wavelet analysis to classify fault types with high accuracy.

Despite their strong interpretability and theoretical rigor, model-based methods face limitations in scalability and applicability to complex, nonlinear, or poorly understood systems. As noted by Chauhan et al. [10], recent research increasingly integrates model-based techniques with data-driven approaches to mitigate these challenges and enhance generalization capabilities.

b) Data-driven methods: Data-driven methods rely on the analysis of historical and real-time sensor data to learn patterns associated with normal and faulty conditions, without requiring explicit physical models of the system. These approaches are particularly advantageous in systems where physical modeling is complex, inaccurate, or unavailable and where extensive sensor datasets can be leveraged to extract meaningful insights [23], [125].

Fig. 9 illustrates the general structure of a data-driven SFD framework. The process begins with sensor data collection, followed by preprocessing steps such as filtering and feature extraction. The processed data are then analyzed using statistical, ML, or DL models to detect anomalies or deviations from expected behavior. Finally, residuals or anomaly scores are evaluated for fault detection, isolation, and estimation.

Statistical methods: Statistical techniques, such as statistical process control, principal component analysis (PCA), and multivariate statistical analysis, detect anomalies by monitoring changes in the characteristics of the sensor signal over time. Yu et al. [126] used improved PCA in conjunction with bond graph modeling for fault diagnosis in redundant actuator systems in aviation. These methods are well-suited for fault detection and sometimes isolation, particularly in high-dimensional data settings. However, their performance

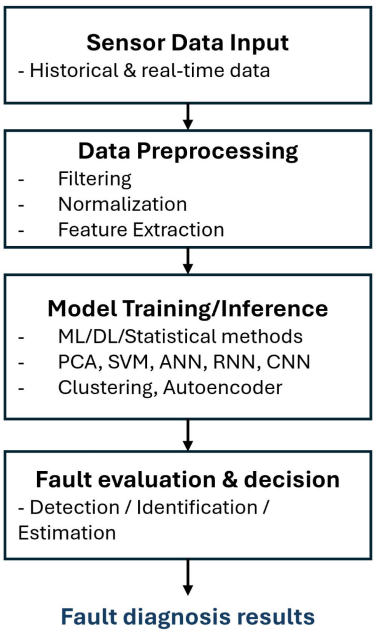


FIGURE 9. Structure of data-driven techniques for SFD.

may degrade under subtle fault conditions or in highly dynamic environments. Similarly, Hou and Ji in [63] proposed a second-order statistical monitoring approach to detect precision degradation by analyzing increases in measurement variance, effectively distinguishing precision loss from other fault modes in industrial processes. The effects of loss of accuracy extend to sensitive applications such as continuous glucose monitoring systems, where increased sensor noise can obscure critical physiological signals, jeopardizing patient safety [62]. Jiang et al. [64] further underscored the importance of early detection of sensor degradation in linear systems to maintain the reliability of control decisions.

ML and DL: ML and DL methods are increasingly being used for SFD due to their ability to learn complex nonlinear relationships directly from data [24], [25]. Supervised learning models, including artificial neural networks (ANNs), SVMs, and decision trees, are trained on labeled datasets to classify faults [18], [127]. For example, Elnokity et al. [18] demonstrated an ANN-based approach capable of detecting, isolating, and estimating sensor faults in a nuclear process. DL architectures such as convolutional neural networks, recurrent neural networks (RNNs), and LSTM autoencoders are used to detect temporal or spatial anomalies in sensor data [128]. Jan et al. [24] proposed a distributed architecture in which edge devices perform detection using autoencoders and central servers manage diagnosis using fuzzy deep neural networks (FDNNs).

These models can support the three diagnostic tasks, detection, isolation, and estimation, but often require significant training data, which is addressed through transfer learning and data enhancement [129]. Unsupervised learning techniques such as clustering and anomaly detection operate without labeled data are useful for detecting novel or rare

fault types. Lundgren and Jung [130] employed open set classification with Kullback–Leibler divergence to manage unknown fault classes and unbalanced data, a common issue in real-world scenarios. Recent research has expanded the use of data-driven techniques in a variety of domains. Lydakis et al. [23] utilized vibration-based sensor data in structural health monitoring to isolate faulty signals using overdetermined systems. Zhao et al. [25] provided a comprehensive review of ML-based SFD for lithium-ion batteries, emphasizing fault tolerance in battery management systems. Darvishi et al. [131] implemented an ML architecture for digital twins, achieving real-time detection and isolation in industrial systems.

In summary, data-driven techniques offer powerful, scalable tools for SFD. Their flexibility and ability to generalize across systems make them suitable for modern, data-rich environments. However, their success depends on the availability, quality, and labeling of the data, as well as ongoing efforts to enhance the interpretability and robustness of the model.

c) Signal-based methods: Signal-based techniques analyze the characteristics of sensor output signals across different domains—time, frequency, and time-frequency—to identify patterns that may indicate faults [132], [133]. These methods are especially effective for detecting dynamic anomalies such as amplitude shifts, transient disturbances, and periodic distortions, making them suitable for early fault detection or integration with ML frameworks [134].

In the time domain, diagnostic algorithms often monitor changes in signal statistics, such as mean, variance, skewness, and kurtosis. In the frequency domain, transformations like the Fast fourier Transform enable the identification of harmonics, sidebands, or spectral imbalances. Time-frequency domain techniques, such as wavelet transforms, higher order spectral analysis, and empirical mode decomposition, capture temporal and frequency information, facilitating the detection of transient, and evolving faults [132], [133]. Fig. 10 illustrates the general workflow for signal-based fault diagnosis. The raw sensor data undergo preprocessing and are transformed into alternative domains. Features such as energy, entropy, spectral peaks, and wavelet coefficients are extracted and passed into a decision mechanism, which could involve rule-based thresholds or a statistical classifier.

Germán-Salló and Strnad [132] used wavelet-based techniques to diagnose electrical drive failures in manufacturing systems. Li et al. [133] demonstrated how a wavelet neural network hybrid can enhance the diagnosis of sensor bias and drift. In the domain of rotating machinery, Kia et al. [134] reviewed envelope analysis and current demodulation techniques to identify faults such as misalignment and bearing defects. Signal-based techniques have also been explored in nonmechanical systems. For example, Wang et al. [135] reviewed infrared thermography as a noninvasive signal-based method for fault diagnosis in rotating machinery. These approaches are particularly valuable for online monitoring due to their speed and sensitivity to subtle signal variations [10], [28]. In short, signal-based techniques are powerful tools

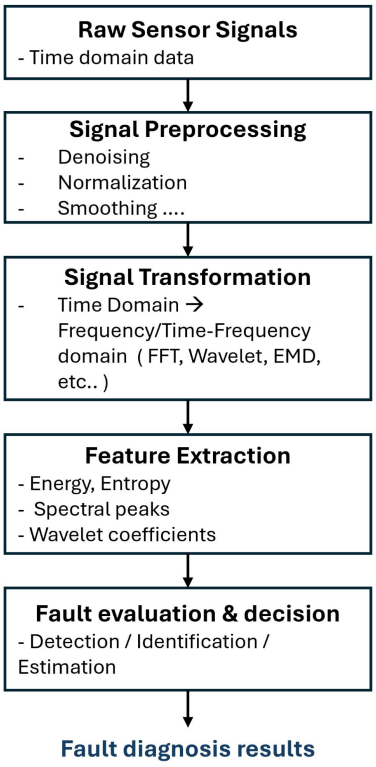


FIGURE 10. Structure of signal-based techniques for SFD.

for diagnosing sensor faults, either on their own or alongside other methods. They are especially useful for detecting faults in complex, nonlinear signals, particularly when physical models are missing or incomplete.

d) Knowledge-based methods: Knowledge-based methods utilize expert knowledge, heuristic rules, and qualitative reasoning frameworks to identify and diagnose sensor faults. Rather than relying on numerical models or large datasets, these approaches encode domain expertise in a structured form, such as rule sets, fuzzy logic systems, and probabilistic graphical models such as Bayesian networks [136], [137]. Expert systems are among the most established tools in this category. They use IF–THEN rules to infer fault conditions from sensor data or derived features. These systems are particularly effective for fault isolation in structured domains with well-documented expert knowledge. Chi et al. [137] emphasized their relevance in industrial IoT settings, where ontologies and rule-based reasoning enable the interpretation of semantic faults in distributed environments. Fuzzy logic is another core technique in this domain, capable of handling imprecision and uncertainty in sensor data. Mazzoleni et al. [138] proposed a fuzzy inference system that integrates expert heuristic knowledge, vibration envelope analysis, and anomaly detection to diagnose faults in industrial machinery. Their results showed robust performance even in the absence of labeled data or precise models. Hybrid approaches further expand the potential of knowledge-based reasoning. TayebiHaghighi et al. [139] designed a hybrid fuzzy observer system that combines Lyapunov-based estimation with SVMs

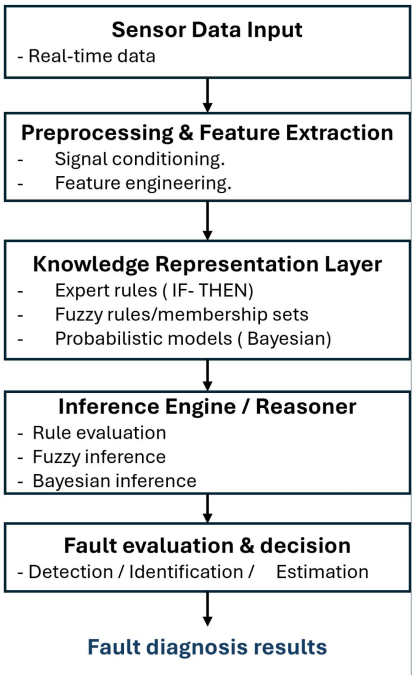


FIGURE 11. Structure of knowledge-based SFD.

to classify internal combustion engine faults, achieving an accuracy of classification greater than 98% . This illustrates the complementary nature of combining symbolic reasoning with data-driven learning. Bayesian networks provide a probabilistic framework for modeling fault relationships and uncertainties. Ademujimi et al. [140] proposed a fusion learning approach that combines quantitative sensor data with qualitative maintenance logs to create richer Bayesian models. Similarly, Iplik et al. [141] showed that Bayesian networks can effectively model fault dependencies in complex sensor networks, enhancing both transparency and diagnostic accuracy. Fig. 11 presents the workflow for knowledge-based fault diagnosis. Sensor data are processed using rule engines, fuzzy systems, or probabilistic inference modules that incorporate domain knowledge, leading to interpretable fault classification and decision making.

Although knowledge-based methods are advantageous in structured environments with known fault mechanisms, their effectiveness can be degraded with novel or complex faults unless continuously updated. To address these challenges, hybrid frameworks that blend knowledge-based reasoning with data-driven or model-based methods are gaining traction [26].

3) DISTRIBUTED AND DECENTRALIZED FAULT DIAGNOSIS STRATEGIES

Traditional SFD techniques often rely on centralized architectures, where all sensor data are gathered in a central processing unit for residual generation, decision making, and diagnosis. Although this approach can offer high diagnostic accuracy for small- to moderate-scale systems, it can quickly become impractical or costly for large-scale spatially distributed sensor networks due to communication overhead, single-point failures, and scalability constraints.

In contrast, distributed and decentralized fault diagnosis frameworks have emerged to address these limitations by shifting part or all of the estimation and detection tasks to the edges of the network. In a distributed setup, multiple local agents (e.g., sensors or nodes) process partial measurements and share information through local communication to collectively detect, isolate, and mitigate faults without requiring a global coordinator [142], [143], [144]. For example, Doostmohammadian et al. [142] proposed a distributed detection and mitigation scheme for biasing attacks over multiagent networks. In this approach, each agent locally generates residuals and uses probabilistic thresholds for attack detection, then applies a system-digraph-based strategy to maintain observability by replacing compromised measurements with observationally equivalent alternatives. Similarly, Doostmohammadian and Meskin [143] presented a networked estimation framework for simultaneous sensor fault detection and isolation in full-rank dynamical systems. Here, a consensus-based observer generates residuals while ensuring network observability, and faulty nodes can be replaced through graph-theoretic sensor replacement. Consensus-based residual generation and decision-making have also been explored by Stankovic et al. [144], who combined local optimal stochastic observers with dynamic consensus to enable overlapping decentralized FDI. This overlapping structure allows for redundancy and robust fault detection, even in partially observable subsystems. To further enhance detection robustness, Reppa et al. [145] developed a two-level distributed fault diagnosis architecture for interconnected CPS. Each subsystem employs a monitoring agent that communicates with neighbors, while a global combinatorial logic resolves ambiguities and isolates multiple faults that might propagate through the network. ML has also been integrated into distributed detection. Jan et al. [24] introduced a hybrid system where lightweight detection blocks run locally at sensor nodes using autoencoders and SVMs, while more computationally intensive fault diagnosis is performed centrally using an FDNN, balancing accuracy with the resource constraints of edge devices. In addition to standard residual-based methods, distributed setups can leverage time-windowed detection using statistical thresholds. Doostmohammadian et al. [146] demonstrated a distributed χ^2 detector for simultaneous estimation and attack isolation in social networks, using Kronecker product network models and block-diagonal gain matrices to enable local agents to detect anomalies independently, even in the presence of unbounded Gaussian noise. In general, distributed and decentralized fault diagnosis strategies improve scalability, resilience to local failures, and fault tolerance in large-scale sensor networks. They represent an important complement to traditional centralized architectures, particularly for IoT, multiagent, and CPS applications where communication cost, security, and local autonomy are critical.

TABLE 9. Capabilities and Tradeoffs of SFD Method Categories

Method Category	Detection	Isolation	Estimation	Core Advantages	Key Trade-Offs
Hardware Redundancy	+++	+++	+	High robustness, immediate detection, simple voting logic	High cost, weight penalty, added hardware complexity; poor scalability for large sensor networks
Model-Based	+++	++	++	Good robustness if model accurate, interpretable residuals, well-suited for physical systems	Requires accurate models, expert design effort; moderate computational burden; limited scalability for highly nonlinear or evolving systems
Data-Driven	+++	++	++	Highly flexible, adaptable, good for complex and nonlinear faults	Needs large training data; high computational cost for training; limited interpretability; scalability depends on data pipeline
Signal-Based	+++	++	+	Low cost, easy to deploy, fast anomaly detection	Low robustness to noise, can misinterpret harmless variations; limited estimation; moderate scalability
Knowledge-Based	++	+++	+	Clear logic, good for known patterns, explainable rules	Poor scalability for unknown faults; rule maintenance overhead; rigid for dynamic systems; moderate robustness only for known conditions

Legend: + = Basic Capability, ++ = Moderate, +++ = Strong Capability.

4) COMPARATIVE ANALYSIS AND TRADEOFFS OF DIAGNOSIS METHODS

The selection of an appropriate SFD strategy depends strongly on the specific application scenario, including factors such as fault criticality, real-time constraints, dominant fault types (e.g., bias, drift, intermittent), the availability of physical models or training data, and the computational resources available [28], [136]. Table 9 summarizes the core capabilities of five representative SFD categories: hardware-based, model-based, data-driven, signal-based, and knowledge-based approaches, while explicitly highlighting their typical tradeoffs in terms of cost, implementation complexity, robustness, and scalability.

This enhanced comparison highlights that each category of methods achieves a distinct balance between diagnostic power and practical tradeoffs in real-world deployment. *Hardware redundancy* is the gold standard for robustness in detection and isolation thanks to direct physical comparison, but its cost, added hardware complexity, and poor scalability make it impractical for large-scale, lightweight, or resource-constrained applications such as IoT and UAVs. *Model-based methods* offer an attractive middle ground: when accurate physical models exist, they deliver strong fault detection and estimation with interpretable residuals. However, they require significant modeling effort and deep domain knowledge, and may struggle with nonlinearity or evolving dynamics, affecting scalability and robustness if model assumptions break down. *Data-driven approaches* stand out for adaptability and nonlinearity handling, ideal when system physics are partially known or too complex to model. They excel at robust detection, but require extensive, high-quality labeled data, raise computational cost, especially for real-time deployment, and pose explainability challenges. Scalability hinges on data availability and infrastructure. *Signal-based techniques* provide simple, cost-effective, and fast anomaly detection, especially for transient or incipient faults. They do not require a detailed model and are easy to retrofit, but robustness suffers under noise, estimation capability is limited, and they often produce false positives if not carefully tuned. *Knowledge-based systems* are strong in isolation for well-known faults thanks to transparent rule-based logic. They are robust for defined conditions and easy to explain to domain experts. However, scalability is poor for unforeseen faults, the maintenance effort increases dramatically with the complexity of the system, and the flexibility is limited for dynamic or novel fault scenarios. *Distributed and decentralized implementations* cut across several categories, especially model-based, observer-based, and redundant architectures, and offer enhanced scalability, robustness, and fault tolerance in large, networked, or multiagent settings. Unlike centralized methods that rely on a single coordinator, distributed strategies leverage consensus, local residuals, and redundant observers to detect and isolate faults cooperatively. These schemes reduce single points of failure and can handle partial observability loss using design concepts such as Q-redundancy and observational equivalence. However, they introduce additional communication overhead, require reliable network connectivity, and their complexity typically grows polynomially with the number of agents and communication links. Despite these challenges, distributed approaches are increasingly attractive for modern IoT, smart grid, and CPS scenarios where system scale and autonomy require flexible and resilient fault management. In general, hardware redundancy and signal-based approaches remain lightweight and near-linear in computational demand,

whereas model-based and data-driven techniques often involve polynomial-time complexity in system size, model dimension, or training data volume, which must be carefully managed, especially for real-time or large-scale deployment. A shared limitation is the reliance on clean baseline data or stable reference behavior, which is often unrealistic for aging or noisy environments. Future research should target hybrid frameworks that combine model-driven rigor, adaptive learning, and scalable architectures to reduce dependency on ideal conditions while balancing cost, robustness, complexity, and real-time performance. Such integrated strategies are vital to enable resilient, explainable, and cost-effective sensing in increasingly complex and autonomous systems.

C. KEY CHALLENGES: NOISE AND FALSE ALARMS

One ongoing challenge in SFD is reliably telling real faults apart from normal measurement noise, especially in dynamic or harsh environments [147], [148]. Although noise is typically modeled as random zero-mean fluctuations (often Gaussian or additive white noise), real-world noise can also include colored, impulse, or flicker noise components that complicate residual analysis [147]. In contrast, sensor faults manifest as systematic deviations such as bias, drift, and freezing that require residual generation schemes that filter the influence of noise while remaining sensitive to faults [148], [149]. Model-based methods such as unknown input observers (UIOs) [149] and adaptive filtering techniques [150] explicitly decouple known disturbances from the fault-sensitive residual. Data-driven and ML methods increasingly address noise by learning typical patterns, leveraging redundancy, and using time-windowed residuals or statistical tests (e.g., χ^2 detectors) to suppress transient noise effects [146], [148]. The assumptions about noise type and fault signatures directly influence the false alarm rate (FAR), the probability of mistakenly declaring a fault when the system is healthy. Many works use statistical hypothesis tests and probabilistic thresholds based on noise variance and distribution [150], [151]. Adaptive threshold algorithms dynamically adjust detection limits using real-time residual statistics to keep the FAR within acceptable limits, even under unexpected disturbances [150]. In networked setups, multiagent consensus or N-out-of-M voting rules further suppress false alarms by validating local alarms across multiple nodes [147], [148]. Despite these advances, balancing high fault sensitivity with low FAR remains an open research challenge, especially for incipient faults masked by high noise. Future solutions must refine robust detection thresholds, hybrid residual filters, and context-aware fusion strategies to enhance fault detectability without sacrificing reliability.

V. STATE-OF-THE-ART OF SENSOR FAULT MITIGATION SOLUTIONS

As systems become increasingly dependent on sensor networks for monitoring, control, and decision making, mitigating the effects of sensor faults becomes a critical concern.

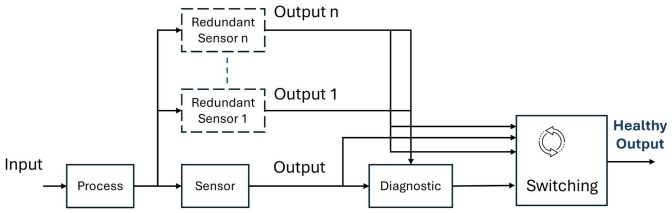


FIGURE 12. Structure of hardware redundancy for sensor fault mitigation.

Sensor faults can degrade system performance, lead to incorrect decisions, or even cause system failures, especially in safety-critical domains such as aerospace, autonomous vehicles, industrial automation, and healthcare [10], [152]. This section provides an in-depth review of current sensor fault mitigation strategies. It goes beyond fault detection and diagnosis to focus on approaches that aim to rectify, compensate, or minimize the impact of sensor faults. These strategies can be preventive, reactive, or adaptive and often involve a combination of hardware, software, and data-driven methodologies [11], [153].

A. HARDWARE REDUNDANCY-BASED APPROACHES

Redundancy has long served as a foundational strategy in sensor fault mitigation, particularly in safety-critical systems such as aviation, automotive control, and biomedical monitoring. *Hardware redundancy* refers to the deployment of multiple physical sensors that measure the same parameter to ensure the continuity of the measurement in the presence of faults [154]. Fig. 12 illustrates a generalized hardware redundancy architecture. The process generates physical input, which is simultaneously measured by a primary sensor and a set of n redundant sensors. These parallel sensor outputs are analyzed by a Diagnostic Unit that compares signal consistency and detects potential faults. Diagnostic analysis informs the *switching unit*, which then selects the healthiest available sensor output, through voting logic or other decision-making strategies, to produce a healthy output signal that ensures the continuity of the system.

A common implementation of hardware redundancy is triple modular redundancy, where three identical sensors are used in parallel, and a majority voting logic determines the correct measurement. This technique effectively masks single-point sensor failures and ensures operational safety. It has been widely adopted in space and avionics applications due to its simplicity and reliability [155], [156]. However, added physical components increase cost, space, weight, and energy consumption—factors that are critical in embedded systems and lightweight platforms. Moreover, hardware redundancy alone cannot handle common-mode failures, where multiple sensors are affected similarly due to shared physical conditions or systemic design flaws [154], [157]. To enhance robustness, hybrid architectures often incorporate observer-based fault detection alongside redundant hardware. Mattei and Paviglianiti [158] proposed integrating H_∞ observers to

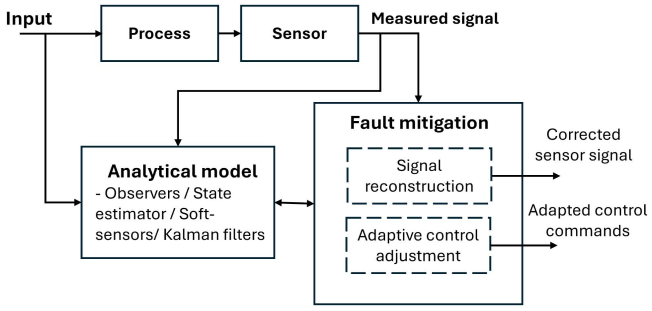


FIGURE 13. Structure of analytical model-based fault mitigation.

reinforce fault isolation capabilities within a hardware redundant aircraft sensor system, demonstrating resilience under noise and model uncertainty. Similarly, Du et al. [156] utilized a distributed Kalman filtering strategy in MEMS gyroscopes, assigning dynamic weights based on signal integrity metrics to fuse sensor outputs. In ultralow-energy applications, such as body sensor networks and serial-triple modular redundant has been introduced to ensure fault tolerance while minimizing power overhead [159]. Furthermore, some frameworks aim to minimize sensor count through strategies *minimum redundancy*, where essential diagnostic resolution is preserved using the fewest redundant elements possible [160]. In summary, hardware redundancy remains an effective yet resource-intensive approach to sensor fault mitigation. Its robustness against random faults makes it ideal for systems that demand high reliability. However, its limitations have driven the exploration of complementary methods, including analytical models and software-based redundancy, to deliver scalable and cost-efficient alternatives in next-generation fault-tolerant systems.

B. ANALYTICAL MODEL-BASED APPROACHES

Analytical model-based fault mitigation techniques rely on detailed mathematical representations of the dynamics of the system to maintain reliable sensor output and system performance in the presence of faults. These models simulate the expected behavior of the process, allowing for the detection of inconsistencies in sensor readings and the reconstruction of accurate signals. Particularly in systems governed by known physical laws, this approach allows for compensation without relying on redundant hardware [153], [161]. Fig. 13 illustrates the structure of a typical analytical model-based mitigation framework. The system process feeds into both the physical sensor and a parallel analytical model, which can include state observers, Kalman filters, or soft sensors. The measured signal is continuously compared to the prediction of the model. When a discrepancy is detected, the fault mitigation unit uses two main strategies: signal reconstruction, which estimates the correct sensor values, and adaptive control adjustment, which modifies control commands to maintain stability and performance. One of the most widely adopted approaches in this domain is the use of state observers. Observers such as Luenberger observers, extended kalman filters (EKF), UKFs, and SMO are instrumental in reconstructing the true state of the system from available measurements [21], [162], [163]. These observers function by continuously comparing predicted outputs from the system model with actual sensor readings, generating residuals that signal inconsistencies caused by faults. For example, Hu et al. [164] proposed a modified EKF framework capable of detecting and compensating gross measurement errors. This technique incorporates statistical testing to identify outliers and biases, which are then quantified and corrected before the observer re-estimates the accurate system state, demonstrating a significant reduction in estimation error. In addition to standard observers, adaptive observers dynamically tune estimation parameters to account for fault-induced deviations in system behavior. Zhang et al. [165] employed an ASMO to handle multiple simultaneous sensor faults in forklift systems. By introducing auxiliary variables and adaptive gains, their approach successfully reconstructed faulty signals in real time, even without prior knowledge of fault magnitudes. Similarly, Dimassi [166] proposed a robust method that combines a high-gain sliding mode and UIOs to simultaneously reconstruct actuator and sensor faults, thereby extending the applicability to systems with relaxed observability assumptions. In control-oriented contexts, adaptive control schemes complement observers by modifying control laws based on estimated fault characteristics. For example, Pazera et al. [21] introduced an adaptive estimator based on H-infinity capable of real-time fault reconstruction while maintaining robustness against disturbances. In high-speed aerospace systems, Hu et al. [167] integrated fault detection, estimation, and compensation within an adaptive sliding-mode control framework, achieving reliable performance under severe fault conditions without the need for separate diagnostic modules. Beyond dynamic observers, soft sensors, or virtual sensors offer an alternative means of fault mitigation in complex systems such as heating, ventilation, and air conditioning (HVAC) or chemical processes. These software-based models replicate sensor behavior using historical or correlated measurements. Nie et al. [168] demonstrated how a dual soft-sensor system could diagnose and accommodate sensor faults in HVAC setups, where predicted values replaced erroneous signals, thus maintaining operational continuity with minimal false alarms. Similarly, Li et al. [169] employed a soft sensor integrated with a Kalman filter bank to reconstruct sensor signals during the acceleration phases of a variable cycle engine, ensuring smooth and reliable control under sensor degradation. Further advancing fault-tolerant capabilities, Du et al. [156] developed a distributed Kalman filtering method for MEMS inertial navigation systems, weighting redundant gyroscope inputs based on a quality index to fuse data adaptively under fault conditions. This technique exemplifies how data fusion and redundancy management enhance mitigation in systems with inherent variability and noise. The efficacy of model-based techniques extends to industrial and energy systems. Oh et al. [170] validated fault-tolerant control methods in proton exchange membrane fuel cell systems, where predictive models were used to maintain performance during sensor failures

through feedback and forward adjustments. In nuclear power plant monitoring, Choi and Lee [171] demonstrated the utility of model-based recurrent neural networks (RNN) frameworks to maintain diagnostic reliability even when the inputs were corrupted.

Although the advantages of model-based mitigation are clear in terms of precision and robustness, their effectiveness depends critically on the accuracy of the underlying models and the availability of reliable system parameters. Moreover, implementation in real-time systems requires careful consideration of computational complexity, particularly for nonlinear systems or applications with stringent latency constraints. However, their ability to deliver quantitative fault compensation and maintain closed-loop performance positions analytical model-based techniques as a cornerstone in advanced fault-tolerant system design.

Recent advances have introduced distributed redundant observer designs that enhance fault tolerance and system survivability in sensor networks. For instance, Doostmohammadian and Charalambous [172] proposed a distributed anomaly detection and estimation approach that employs Q-redundant observers. This design ensures that even if up to Q sensors fail or are removed, the system maintains observability by leveraging observational equivalence within the network. Such distributed frameworks allow local agents to recover lost information and preserve estimation performance without a central coordinator. Similar ideas appear in recent works using chi-square detectors and local residual tracking to improve detection robustness under stochastic disturbances and adversarial scenarios [146], [173]. By combining local residual history with networked estimation, these methods enable scalable, survivable sensor fault mitigation suitable for large-scale and mission-critical CPS.

C. AI-BASED APPROACHES

AI-based fault mitigation techniques have emerged as powerful solutions to enhance sensor reliability in complex and data-rich environments where traditional model-based methods may fail. These approaches utilize ML, DL, and reinforcement learning (RL) to detect, classify, estimate, and compensate for sensor faults, thus improving system resilience without relying on explicit physical models. As illustrated in Fig. 14, the pipeline includes sensor data collection, preprocessing, inference of the AI model (ML, DL, RL), and fault rectification to produce healthy corrected sensor data. The mitigation process begins with the input of sensor data, incorporating historical datasets and real-time streaming data. These raw data undergo essential preprocessing steps such as filtering, normalization, and feature extraction to improve signal quality and reduce noise. The core component is the AI inference engine, where various models are trained and applied. ML algorithms such as SVM and random forests are effective for fault classification and signal reconstruction based on statistical patterns in labeled datasets [58], [174]. DL methods, including repeat neural networks (RNNs), LSTM networks, and autoencoders, are particularly suited for capturing

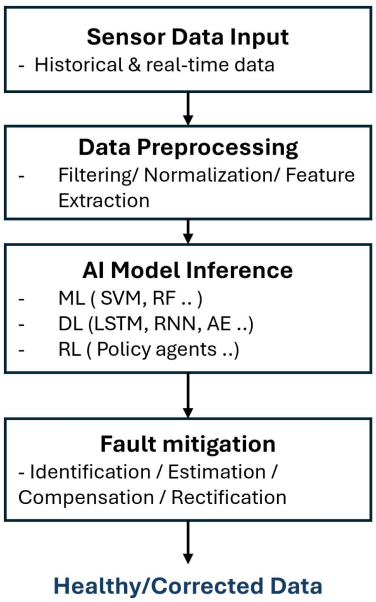


FIGURE 14. General architecture of AI-based fault mitigation.

temporal dependencies and complex nonlinearities, allowing high-fidelity signal reconstruction even under severe drift or missing data conditions [19], [175], [176], [177]. In parallel, RL techniques allow agents to learn optimal fault compensation strategies through interaction with the system, guided by reward functions that capture system performance, safety, or stability criteria [178], [179], [180]. The output of these AI models feeds into a fault mitigation module that performs key tasks, such as identification, estimation, compensation, and rectification, ultimately producing healthy and corrected data streams. These data can be reintegrated into the control system, ensuring continued operation and fault tolerance. Despite challenges such as data dependency, generalization to novel fault types, and explainability, AI-based methods are increasingly being integrated into hybrid frameworks and digital twin architectures to overcome these limitations. Their ability to scale, adapt, and operate autonomously positions them as critical tools in the next generation of intelligent, fault-tolerant systems.

D. CALIBRATION AND MAINTENANCE APPROACHES

Preventive fault mitigation techniques grounded in sensor calibration and predictive maintenance form the first line of defense against sensor degradation. Rather than reacting to faults postoccurrence, these strategies aim to continuously preserve measurement fidelity and system reliability through preemptive actions. Although closely related in their preventive intent, calibration, and maintenance rely on different mechanisms and are best considered as complementary, parallel strategies [181], [182]. Sensor calibration focuses on correcting systematic sensor errors, such as bias, drift, and scale factor changes, through automated adaptive routines. These methods are often embedded within the sensor or control architecture and rely on a comparison with known references or internal model consistency checks [183], [184].

TABLE 10. Comparative Summary of Sensor Fault Mitigation Approaches

Mitigation Strategy	Key Advantages	Key Challenges	Typical Trade-Offs
Hardware Redundancy	High reliability; robust masking of faults; simple voting logic	Costly in hardware, weight, and energy; limited scalability for large or weight-sensitive systems	Robustness vs. cost and size
Analytical Model-Based	Quantitative fault compensation; suitable for predictable dynamics; low added hardware	Requires accurate models; sensitive to uncertainties; higher computational burden for nonlinear systems	Accuracy vs. modeling effort and complexity
AI-Based Methods	Adaptive, data-driven; capable of handling complex patterns and nonlinearities; improves over time	Needs large diverse datasets; can be opaque; may struggle with explainability and trust	Flexibility vs. data/compute demands
Calibration and Maintenance	Preserves baseline performance; cost-effective over sensor lifetime; prevents degradation proactively	Requires periodic intervention or robust auto-calibration algorithms; limited real-time fault handling	Reliability vs. maintenance cost and scheduling

For example, Jia et al. [77] proposed a closed-form calibration framework that optimizes the sensor configuration by leveraging geometric relationships and correction of measurement bias. In decentralized systems such as wireless sensor networks, Butt et al. [185] applied an EKF algorithm with beacon-based localization to achieve continuous self-calibration under noisy conditions. Noncontact voltage sensors, similarly, use dynamic capacitance adjustments to maintain calibration across variable configurations [186]. Furthermore, data-driven models such as the neural network approach of Kim et al. [187] offered robust adaptive calibration, particularly in HVAC applications where real-time fault correction is necessary without physical intervention. Predictive maintenance, on the contrary, emphasizes early anomaly detection and health monitoring rather than direct calibration. These techniques use operational health indicators, such as noise levels, data drift, and power consumption, to predict imminent sensor failures. Maintenance is then triggered on the basis of the condition, minimizing downtime and unnecessary intervention [51]. For example, Zhengwei and Gongsheng [188] introduced a framework that prioritizes sensor maintenance based on the criticality to system performance and predicted degradation. Martins et al. [189] implemented hidden Markov models to detect deviations in calibration performance, triggering corrective actions only when statistically warranted. Similarly, Mazzoli et al. [190] developed a self-diagnostic and self-compensating module for displacement sensors that drastically improved accuracy despite sensor wear, illustrating the practical benefits of condition-based maintenance in aging systems.

Fig. 15 illustrates the calibration pathway, where sensor inputs undergo automated adjustments and model updates to ensure continuous accuracy. while Fig. 16 presents the predictive maintenance process, in which sensor data is evaluated for anomalies, and preventive actions are triggered through a condition-based update mechanism. Both pathways ultimately contribute to reliable and healthy sensor data output.

Together, these strategies play an important role in maintaining sensor reliability and ensuring fault-resilient system operation. Their integration with ML, adaptive filtering, and model-based estimation techniques continues to evolve, offering enhanced scalability for large-scale, autonomous, or safety-critical systems.

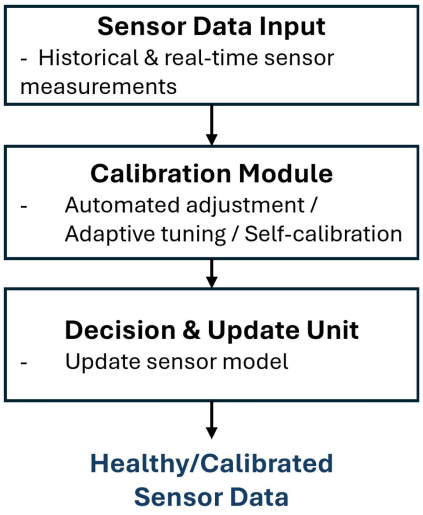


FIGURE 15. General architecture of Calibration-based fault mitigation.

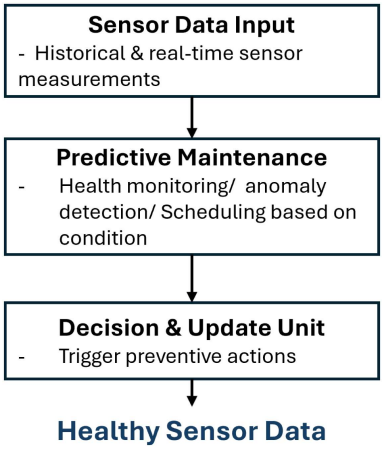


FIGURE 16. General architecture of Maintenance-based fault mitigation.

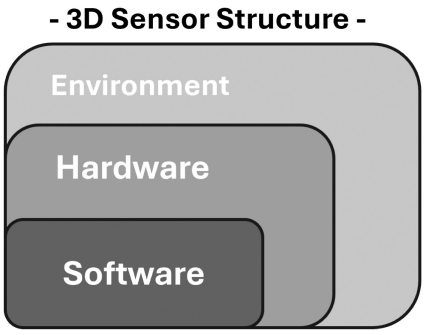


FIGURE 17. Three-dimensional sensor structure.

E. COMPARATIVE ANALYSIS AND DISCUSSION OF SENSOR FAULT-MITIGATION SOLUTIONS

The range of sensor fault mitigation strategies reflects the diverse demands of modern engineering systems. Table 10 summarizes the main advantages, challenges and typical tradeoffs of hardware redundancy, analytical model-based, AI-driven, and calibration/maintenance solutions.

This comparative view shows that each class tackles sensor faults from a different angle: No single approach can fully address all challenges alone. Hardware redundancy excels in fault masking, but scales poorly. Analytical and AI-based techniques offer dynamic compensation, but depend on models or data availability. Calibration and predictive maintenance extend the life of the sensor and prevent faults, but do not directly handle abrupt failures. In practice, hardware redundancy and calibration-based strategies impose minimal computational burden but increase physical or operational cost, whereas analytical and AI-based mitigation methods generally require higher computational resources for real-time estimation and adaptive compensation, especially in large-scale or resource-constrained systems. As systems grow in complexity and autonomy, future research should integrate these strategies into hybrid, cost-effective, and self-adaptive frameworks. Such combinations can balance robustness, cost, computational complexity, and scalability to maintain reliable sensing in critical applications.

VI. CONCLUSION

This article presents a detailed analysis of SFD and mitigation strategies, with a focus on their application in intelligent and safety-critical systems. Based on our findings, a key insight is that a sensor should not be treated as a single-layer component. Instead, it should be viewed as a three-dimensional (3-D) system composed of software, hardware, and the operating environment. As shown in Fig. 17, faults can originate in any of these layers and propagate through them, affecting the integrity of sensor data. This propagation can affect not only the sensor output but also the performance of the connected subsystems that rely on these data.

One major observation is that most existing fault detection and diagnosis (FDD) frameworks do not take a truly holistic approach. They typically focus only on analyzing the sensor's output signal without explicitly considering whether the observed deviation is caused by hardware degradation, software anomalies, environmental noise, or even external attacks. This makes fault decoupling from both stochastic noise and intentional cyber-interference a persistent and unsolved research problem. Distinguishing between these causes remains an open research question, especially in real-time systems where fast and accurate decisions are critical.

Another important issue is how vulnerable sensors are to noise and external disturbances. Many current FDD methods find it difficult to tell a real sensor fault apart from a temporary deviation caused by external factors. Although adaptive filtering and statistical models provide partial solutions, the field still lacks universally robust decoupling strategies capable of maintaining diagnostic fidelity under operating conditions. Just as important is the challenge of differentiating sensor faults and cyberattacks, which can both result in misleading or corrupted data. Here, the boundary between cybersecurity and sensor health monitoring is increasingly blurred, which calls for interdisciplinary approaches that integrate security measures with physical fault detection.

Another recurring issue is the limited generalizability of existing FDD methods. Many frameworks are tailored to specific sensor types, domains, or operating conditions, which restricts their scalability and drives up the cost of adaptation when deployed in diverse applications. Promising directions, such as domain adaptation, transfer learning, and distributed detection frameworks offer new pathways for more universal solutions, but they require further work to balance adaptability, interpretability, and computational cost.

From the mitigation prespective, our review highlights that the effectiveness of fault mitigation strategies is closely coupled with the speed and precision of the FDD process. An accurate and timely diagnosis is a prerequisite for the successful application of mitigation actions. Recent research trends have shifted from hardware redundancy, which, although highly reliable, introduces cost, complexity, and weight, to analytical signal compensation. These analytical approaches, while promising, are still challenged by limitations in real-time performance and diagnostic accuracy. In addition, a faulty sensor may indicate deeper system-level degradation, increasing the risk that redundant sensors will soon fail, as well. This observation reinforces the need for more advanced, adaptive, and fault-resistant signal reconstruction techniques.

In summary, this review highlights the following key directions for future research.

1) *Unified sensor health frameworks:* Develop integrated FDD architectures that jointly consider hardware degradation, software anomalies, and environmental impacts to enable more accurate, context-aware, and explainable diagnostics.
2) *Robust fault-noise decoupling:* Design advanced residual generation, adaptive thresholding, and multisensor consensus mechanisms to reliably separate true faults from stochastic noise, disturbances, or malicious intrusions, minimizing false alarms.
3) *Scalable and generalizable solutions:* Promote the use of domain adaptation, transfer learning, and hybrid mode and data driven approaches to build diagnostic and mitigation strategies that remain effective across

different sensor types, applications, and evolving operating conditions.

4) *Decentralized and resilient mitigation:* Integrate distributed fault detection with survivable mitigation strategies, such as redundant observer design, to maintain observability and system performance even under multiple simultaneous faults or partial network failures.
5) *Practical real-time implementation:* Develop solutions that balance diagnostic accuracy, computational complexity, and communication costs to support large-scale, real-time deployments in dynamic and safety-critical environments.

The path toward truly reliable and intelligent sensor systems requires bridging the gaps between fault diagnosis, cybersecurity, and adaptive learning. We hope that this review serves as a stepping stone for researchers and engineers aiming to design the next generation of resilient sensor systems capable of operating safely and autonomously in increasingly complex environments.

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